PROMOTION EFFECTIVENESS FOR A RETAIL CHAIN

Promotions lift sales by about 38%. Keep them, fix type b, and get the cost data.

KEEP

Types a, c and d

35-38% uplift across 1,098 of the chain's 1,115 stores.

REDESIGN AND TEST

Type b

About 20% uplift in 17 stores. Only about 1% of the value, so test cheaply.

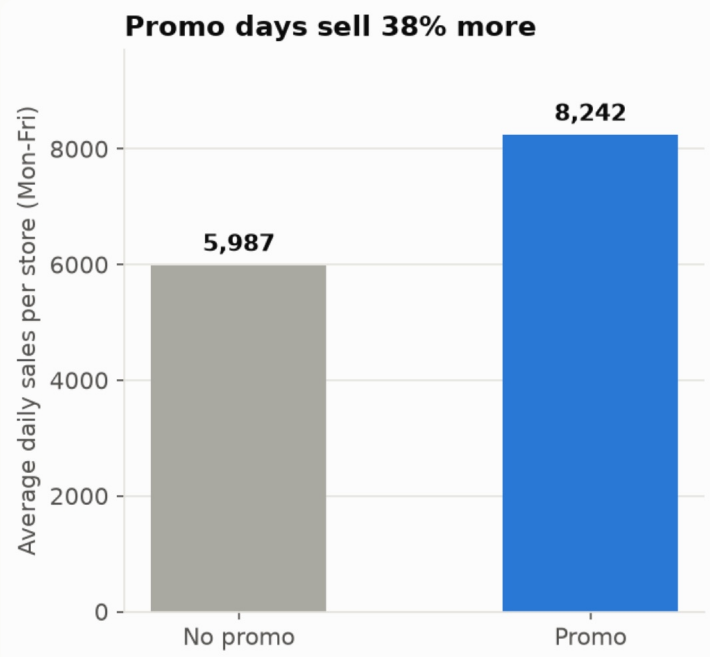
STOP AS A RULE

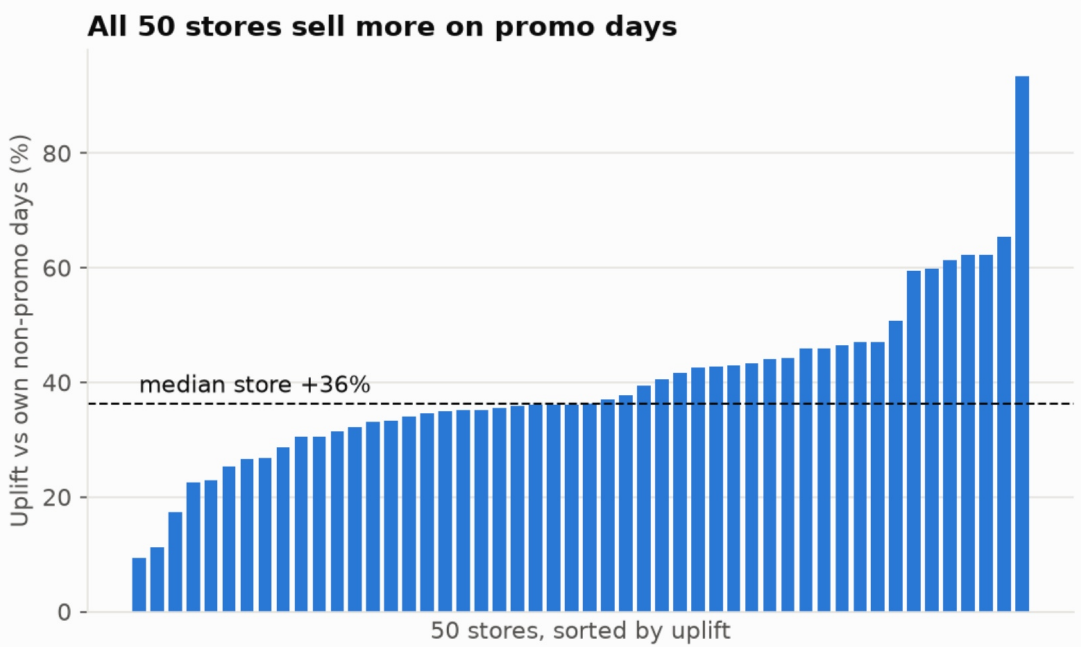
Targeting by competitor distance

Promotions lift sales by a similar amount at every distance.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.

All 50 stores sell more on promo days





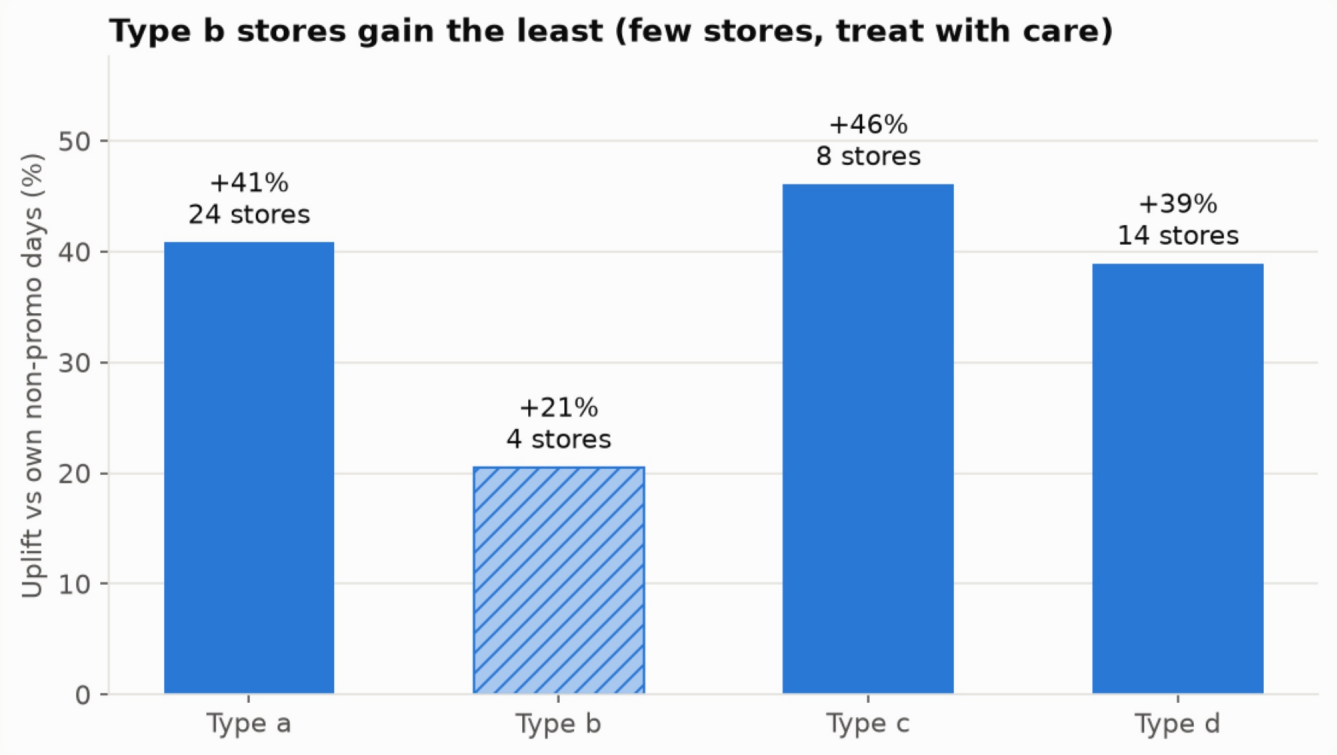
Three methods agree: +35% (model), +38% (store vs itself), +39% (regression).

The lift is more customers (+19%) and a bigger spend per customer (+15%).

Not driven by a few stores: the weakest gains +9%, the median store +36%.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.

Type b gains least, and it is a small group



About +21%

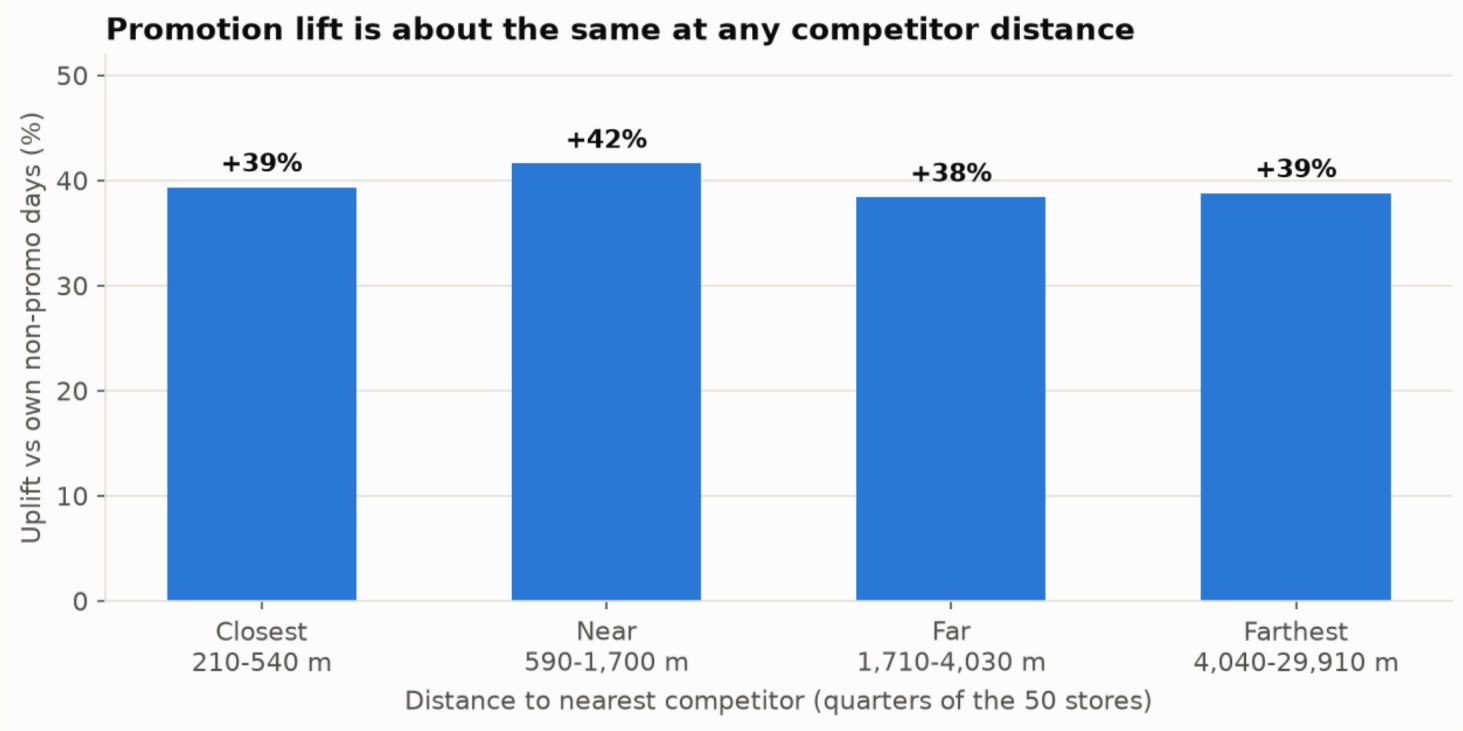
average uplift for type b, against 39-46% for types a, c and d.

Handle with care: only 4 sample stores (17 in the chain). A signal to verify, not a proven fact.

So what: redesign and test type b cheaply.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.

Competitor distance makes no difference



+38-42%

uplift in every distance band.

A store with a competitor 200 m away responds like one with a competitor 10 km away.

So what: do not target promotions by nearby competition.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.

About EUR 330M a year of extra revenue, almost all from types a, c and d

Segment	Stores in chain	Decision	Model uplift	Extra sales per promo day	Break-even cost per promo day*
Types a, c, d	1,098	Keep	35-38%	EUR 2,060-2,415	EUR 410-480
Type b	17	Redesign and test	About 20%	About EUR 1,430	About EUR 290
Targeting by competitor distance	All	Stop as a rule	33-36% at every distance	n/a	n/a

Revenue, not profit. The data has no promotion cost or margin. *Break-even assumes a 20% gross margin, an assumption to replace with the real figure.

EUR 330M is an order of magnitude: one 6-week summer window from 50 stores, scaled to 1,115 stores and a year.

Type b is about 1% of the value. Matching the other types would add up to about EUR 1.9M a year.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.

Confirm cause and profit before scaling

Biggest risks

Not randomised. Part of the lift may be days already expected to be busy: association, not proven cause.

No cost or margin data. We show extra sales, not profit. A stop needs the cost figure.

Small sample. 50 stores; type b rests on only 4.

One short window. The model test covers 6 summer weeks.

Next steps

1. Get promotion cost and margin, and replace the assumed 20% margin.

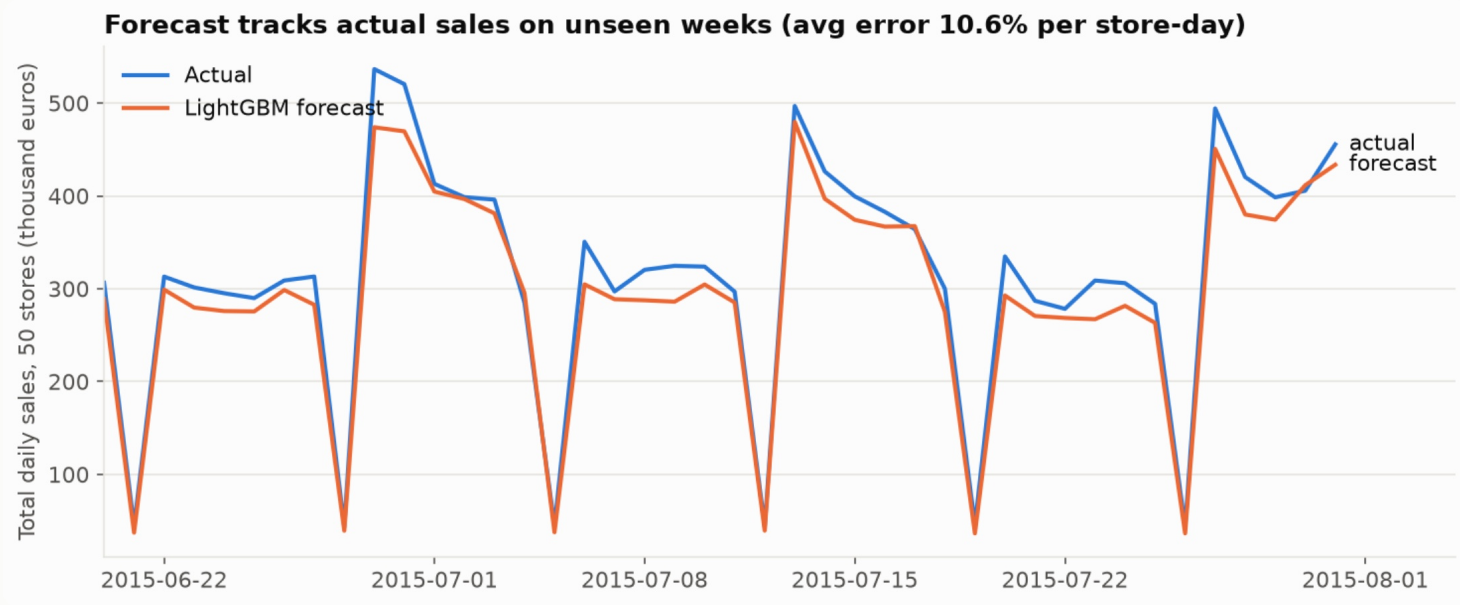
2. Run a controlled test: withhold or shorten promos in matched stores for 6-8 weeks, starting with type b.

3. Repeat on all 1,115 stores and a full year to check seasonality and the type b result.

4. Keep the dashboard as the monthly view of uplift by segment.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.

Appendix: the forecast tracks sales but runs about 6% low



10.6% average error per store-day on 6 weeks the model never saw.

Comparable to, not better than, a simple store x weekday x promo average (10.1%).

The model's value here is the **promo uplift estimate**, not forecasting accuracy.

Source: Rossmann Store Sales (Kaggle), 50-store sample. Sales are revenue, not profit.